

Exam MA-224

Course code: MA-224

Course name: Discrete mathematics 2

Date: 1. December 2025

Time: 09:00 – 13:00 (4 hours)

Number of pages: 8

Number of tasks: 12

Permitted equipment: All written or printed material. Calculator without scripts.
No communication is allowed during the exam.

Remarks: Present your answers so that it is clear what methods you have used, and how you have used them. Your answer should contain a calculation or an argument explaining how the answer was reached; otherwise, it is incomplete.
All problems 1-12 are scored equally. If a problem has more than one part, then each part is scored equally within the problem.

Problem 1

We consider the following theorem.

$$\text{Th1: } [(\neg p \vee \neg q) \implies (p \wedge q \wedge r)] \iff [p \wedge q]$$

- (a) Prove Th1 using truth tables.
- (b) Prove Th1 using logical inferences.

For your convenience, here are some possible inference rules.

double negation: $\neg\neg A \equiv A$

idempotence of $\wedge$: $A \wedge A \equiv A$

commutativity of $\wedge$: $A \wedge B \equiv B \wedge A$

associativity of $\wedge$: $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$

deMorgans law for $\wedge$: $\neg(A \wedge B) \equiv \neg A \vee \neg B$

conjunctive simplification: $(A \wedge B) \implies A$

idempotence of $\vee$: $A \vee A \equiv A$

commutativity of $\vee$: $A \vee B \equiv B \vee A$

associativity of $\vee$: $(A \vee B) \vee C \equiv A \vee (B \vee C)$

deMorgans law for $\vee$: $\neg(A \vee B) \equiv \neg A \wedge \neg B$

disjunctive amplification: $A \implies (A \vee B)$

contraposition: $A \implies B \equiv (\neg A \vee B)$

sylogism: $[(A \implies B) \wedge (B \implies C)] \implies (A \implies C)$

disjunctive syllogism: $[(A \vee B) \wedge \neg B] \implies A$

definition of $\iff$: $[(A \implies B) \wedge (B \implies A)] \equiv (A \iff B)$

Problem 2

Let A be a non-empty set of letters (an alphabet). Strings are defined as follows.

- The empty string ϵ is a string: $\epsilon \in \text{Strings}$.
- If s is a string, and $a \in A$ is a letter, then $s \circ a$ is a string:
 $s \in \text{Strings} \implies s \circ a \in \text{Strings}$.
- There are no other strings.
- If a set S contains ϵ and for each string s and each letter a also $s \circ a$, then S contains all strings.

We consider the following definitions of l (length) and D (double).

1.1: $l(\epsilon) = 0$

1.2: $\forall s \in \text{Strings} : \forall a \in A : l(s \circ a) = l(s) + 1$

D.1: $D(\epsilon) = \epsilon$

D.2: $\forall s \in \text{Strings} : \forall a \in A : D(s \circ a) = (D(s) \circ a) \circ a$

Prove the following theorem by induction using the definitions above and all knowledge you have about integers.

ThD1: $\forall s \in \text{Strings} : l(D(s)) = 2 \cdot l(s)$

Problem 3

Consider the following grammar G , given as a four-tuple (terminals, nonterminals, start symbol, rules).

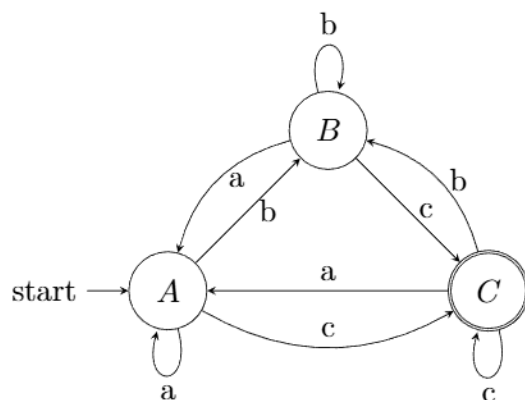
$$G = (\{a, b, c\}, \\ \{A, B, C\}, \\ A, \\ \{A \rightarrow a, A \rightarrow Aa, A \rightarrow B, A \rightarrow BB, \\ B \rightarrow b, B \rightarrow Bb, B \rightarrow C, B \rightarrow CC, \\ C \rightarrow c, C \rightarrow CC\} \\)$$

- (a) Find a derivation for the string "cc".
- (b) Find all syntax trees for the string "cc" and determine whether the grammar is ambiguous.
- (c) Does the ambiguity change when you use a different start symbol?

Give reasons.

Problem 4

Translate the following NFA to a regular expression. Show the steps of the translation. You can simplify the regular expression, if possible.



Problem 5

Consider the following specification, where the operator `mod` calculates the division remainder.

$\{x \geq 30\} \ x, y := x', y' \mid y' \geq 0 \text{ and } y' < x \text{ and } 0 = (x - y') \bmod 3$

For each of the four codes below, check: Is the code a correct implementation of the specification? If yes, how can you use the refinement calculus to prove the correctness? You do not need to do the refinements, just name the relevant refinement rules. If no, provide an example where the code behaves differently from the specification.

- (a) Code: $\{x \geq 0\} \ y := y' \mid y' = x - 3$
- (b) Code: $\{x \geq 3\} \ y := y' \mid y' = x - 3$
- (c) Code: $\{x \geq 0\} \ y := y' \mid y' = x \bmod 3$
- (d) Code: $\{x \geq 3\} \ y := y' \mid y' = x \bmod 3$

You can consider the following refinement rules.

Introduce local variable (InV): $\{\text{pre}\} v:=v' | \text{rel} [= \text{VAR } l:T.l:=E; \{\text{pre}\} v,l:=v',l' | \text{rel}$

Introduce constant (InC): $\{\text{pre}\} v:=v' | \text{rel} [= \text{CONST } c:T.c:=E; \{\text{pre}\} v:=v' | \text{rel}$

Introduce assertion (InA): $E \text{ does not mention } v \implies v:=E [= v:=E; \{v=E\}$

Insert leading assignment (Las): $\text{rel does not mention } x \implies$

$$v,x:=v',x' | \text{rel} [= x:=E; v,x:=v',x' | \text{rel}$$

Strengthen postcondition (Str): $(\text{pre} \wedge \text{rel2} \implies \text{rel1}) \implies$

$$\{\text{pre}\} v:=v' | \text{rel1} [= \{\text{pre}\} v:=v' | \text{rel2}$$

Contract frame (Fra): $x,v:=x',v' | \text{rel} \wedge x'=x [= v:=v' | \text{rel}[x' \setminus x]$

Introduce assignment (Ass): $x:=x' | x'=E [= x:=E$

Rewrite (Rew): $\{E1=E2\} v:=v' | \text{rel}[E1] [= v:=v' | \text{rel}[E2]$

Weaken precondition (Wep): $(\text{pre1} \implies \text{pre2}) \implies \{\text{pre1}\} [= \{\text{pre2}\}$

Introduce IF (InI):

$$\{\text{pre}\} v:=v' | \text{rel} [= \text{if } G \text{ then } \{G \wedge \text{pre}\} v:=v' | \text{rel} \text{ else } \{\neg G \wedge \text{pre}\} v:=v' | \text{rel}$$

Problem 6

Let R be a relation on natural numbers which is true if $a + b$ is even.

$$aRb \equiv 2|(a + b)$$

- (a) Is R reflexive?
- (b) Is R symmetric?
- (c) Is R anti-symmetric?
- (d) Is R transitive?
- (e) Is R an equivalence relation?
- (f) Is R an order (partial or total)?

Give reasons.

Problem 7

For each integer $n \geq 0$, let

$$c_n = (3 + \sqrt{13})^n + (3 - \sqrt{13})^n.$$

Find a second order, linear, homogeneous recurrence relation with non-zero constant coefficients for which c_n is a solution.

Problem 8

Let $G = \{x, y, z, w\}$ be a set consisting of four elements. Any binary operation $\cdot$ on G gives rise to a multiplication table

	x	y	z	w
x	$x \cdot x$	$x \cdot y$	$x \cdot z$	$x \cdot w$
y	$y \cdot x$	$y \cdot y$	$y \cdot z$	$y \cdot w$
z	$z \cdot x$	$z \cdot y$	$z \cdot z$	$z \cdot w$
w	$w \cdot x$	$w \cdot y$	$w \cdot z$	$w \cdot w$

On the other hand, any multiplication table defines a binary operation on G . We say that a multiplication table *corresponds to a group* if it defines a binary operation $\cdot$ such that $(G, \cdot)$ is a group.

Consider the following multiplication tables:

(1)

	x	y	z	w
x	x	y	z	w
y	y	z	z	x
z	z	w	z	y
w	w	x	y	z

(2)

	x	y	z	w
x	y	x	w	z
y	x	y	z	w
z	w	z	y	x
w	z	w	x	y

(3)

	x	y	z	w
x	x	y	z	w
y	y	w	x	z
z	z	x	w	y
w	w	z	y	x

Exactly two of the tables above correspond to groups, and one of them does not correspond to a group. You are free to use this fact without proof in your answers.

- Identify the multiplication table that does not correspond to a group.
- For each table that corresponds to a group, find the neutral element and compute the order of the element w .
- Find all tables that correspond to a cyclic group.

Problem 9

Let p, q be prime numbers such that $p \neq q$. Consider the group $(P_n, \cdot_n)$, where $n = pq$. Assume that we choose an integer x in the range $[1, n]$. Then x may or may not be an element of P_n .

Argue why the probability of having $x \in P_n$ is close to 1.0 when p and q are large prime numbers. (Hint: Compute the order of P_n .)

Problem 10

Let $(G, \cdot)$ be a finite group, with subgroups $H \subset G$ and $K \subset G$. Assume that the order of H is equal to 2, and that the order of K is equal to 8.

Consider the following statements:

- a) The order of G must be an even number.
- b) The order of G must be at least 8.
- c) The order of G must be at least 16.

Identify which of the above statements are true, and which are false. In each case, prove your answer.

Problem 11

The encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 6}$$

is given by matrix multiplication

$$E(w) = w \cdot G,$$

where G is the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

A message is encoded and sent over a noisy channel. A transmission error of weight 2 occurs. Is the recipient guaranteed to be able to detect this error? Remember to explain your answer. In case the answer is “no”, give an example of a message m and an error pattern e of weight 2 such that $r = m + e$ is a codeword.

Problem 12

Given the following generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

The encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

is given by matrix multiplication by G :

$$E(w) = w \cdot G.$$

A message m is encoded, transmitted over a noisy channel, and received as the bitstring $r = 11110$.

Compute the *syndrome* of r and determine if r is a *codeword* for the encoding E .